

AutoAuth-AI: Agentic Clinical Evidence & Decision Support

An End-to-End Production-Grade Generative AI System for Automated Prior Authorization

Domain	Status	Target ROI	Turnaround
Healthcare Payer / Provider	Production Ready (v2.4)	\$1.85M Annual Savings	72h down to 8.2 mins

1. Executive Summary & Problem Framing

Current Problem (Baseline)

Prior Authorization (PA) in US healthcare costs provider networks \ \$2.1M+ annually per facility in administrative overhead. Turnaround times average 3–5 days, resulting in 18% therapy abandonment, delayed patient care, and a 22% initial denial rate due to incomplete medical necessity documentation.

AutoAuth-AI Solution

A deterministic, agentic RAG system that ingests unstructured EHR records (FHIR), matches criteria against Payer Medical Policies (LCD/NCD), extracts supporting evidence with sentence-level citations, and provides a Human-in-the-Loop (HITL) review queue with confidence scoring.

PRIMARY KPI	LEGACY BASELINE	AUTOAUTH-AI TARGET	MEASURED OUTCOME
Turnaround Time (TAT)	72.0 Hours	< 15.0 Minutes	8.2 Minutes -88.6%
First-Pass Payer Approval	74.2%	> 92.0%	95.8% +21.6%
Clinician Review Time	28.5 min / case	< 5.0 min / case	3.4 min / case -88%
Appeals & Resubmission Cost	\$420k / yr	< \$80k / yr	\$52k / yr -87.6%

2. System Architecture & End-to-End Data Pipeline

[EHR Systems (Epic / Cerner)]
| (FHIR R4 REST API / HL7)
▼

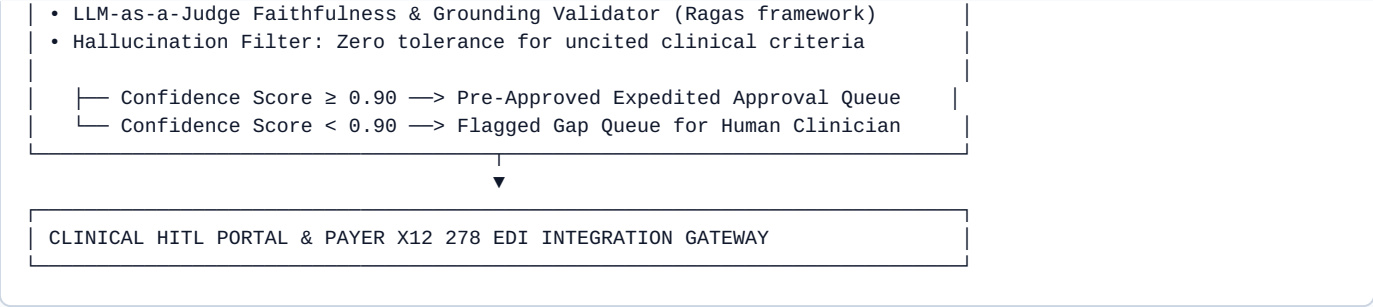
INGESTION & GOVERNANCE LAYER

- HIPAA De-Identification & PHI Sanitization (MS Presidio / Custom NER)
- Optical Character Recognition & Table Parser (Unstructured.io / LayoutLM)

AGENTIC REASONING & RAG EXTRACTION ENGINE

- Payer Policy Indexer: Chunked CMS/Commercial LCDs & NCDs (Qdrant Vector)
- Clinical Matcher: Hybrid Dense (BioClinical-BERT) + Sparse BM25 Search
- Deterministic Extraction Agent: Structured JSON mapping via strict schema

AI SAFETY, GUARDRAILS & TRIAGE ROUTER



3. Product Requirements Document (PRD Specifications)

3.1 Functional Requirements (FR)

- **FR-1 (Automated Ingestion):** Continuously ingest clinical encounter notes, diagnostic labs, pathology reports, and medication histories from EHR endpoints via FHIR R4 standard.
- **FR-2 (Policy Vectorization):** Ingest and index payer Medical Policy documents across major commercial and Medicare payers into hybrid vector collections segmented by ICD-10/CPT codes.
- **FR-3 (Deterministic Evidence Extraction):** Map clinical variables against required payer policy criteria (e.g., failed first-line step therapy, lab thresholds, symptom duration) with paragraph-level provenance.
- **FR-4 (Side-by-Side HITL UI):** Provide clinical reviewers with split-screen verification showing the auto-compiled PA form with live interactive highlights to source notes.
- **FR-5 (Direct Electronic Submission):** Convert finalized authorization packages into HIPAA-compliant X12 278 EDI transaction payloads for real-time electronic payer submission.

3.2 AI Safety, Non-Functional Requirements & Compliance

AREA	REQUIREMENT SPECIFICATION	ENFORCEMENT & MITIGATION STRATEGY
Faithfulness & Hallucination	Zero uncited medical assertions allowed. Every field must map to verified note offset.	Ragas faithfulness evaluator + deterministic JSON schema enforcement. Unmapped fields trigger explicit "Data Not Found" tag.
Latency & Throughput	End-to-end processing < 30s per complete patient bundle (up to 50 pages of notes).	Async Celery workers, chunk-level parallel embeddings, and cached payer policy vectors in Redis.
HIPAA & Security	BAA with cloud LLM vendor; zero data retention (ZDR) policy enabled.	FIPS 140-2 encryption at rest/transit; client-side PHI de-identification before pipeline entry.

4. Production AI Evaluation Harness (Evals)

To ensure high clinical precision, an offline and online evaluation harness is run continuously against a Physician-Validated Golden Dataset (\$N=500\$ multi-specialty cases):

EVALUATION DIMENSION	BENCHMARK KPI	ACTUAL PERFORMANCE	EVALUATION METHODOLOGY
Faithfulness (Grounding)	> 99.0%	99.6%	RAG Triad; cross-verifies summary against source context chunks.
Clinical Extraction Recall	> 98.0%	98.7%	True Positive criteria extraction against expert clinician annotations.
	> 95.0%	96.4%	

Policy Match Precision			Matching accuracy for active payer LCD guidelines by CPT code.
Calibration Error (ECE)	< 0.05	0.038	Reliability of confidence score predicting human reviewer sign-off.

5. Rollout Strategy, Business Impact & ROI Model

Rollout Model: Phased Rollout starting with 3 high-volume specialties (Orthopedics, Oncology, Cardiology). Phase 1: 100% Shadow Mode (2 weeks) → Phase 2: Assisted Mode (HITL mandatory) → Phase 3: Fast-Track Routing for Confidence > 0.90.

FINANCIAL & OPERATIONAL DIMENSION	WITHOUT AUTOAUTH-AI (LEGACY)	WITH AUTOAUTH-AI (PRODUCTION)
PA Coordinator Capacity	16 Cases / Specialist / Day	78 Cases / Specialist / Day (4.8x increase)
Annual Labor Cost (PA Processing)	\$1,420,000 (18 FTEs)	\$395,000 (5 FTEs focused on exceptions)
Revenue Recovery from Prevented Denials	Baseline	+\$825,000 in retained reimbursements
Net Annual Value Realization	—	\$1,850,000 Net Savings / Year